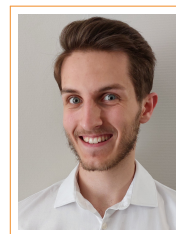


Noé Demange

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Education

- 2024–Present **PhD in Computer Science**, title : **Path Generation for the Synthesis of Molecular Cages**, *DAVID Lab., University of Versailles Saint-Quentin-en-Yvelines / Paris-Saclay*, Versailles, France, theses.fr/s402263.
Supervisors: Yann Strozecki, Sandrine Vial
- 2022–2024 **Master's Degree in Algorithms and Modeling at the Interface of Sciences (AMIS)**, *University Paris-Saclay*, Versailles, France.
Coursework: molecular modeling, graph theory, artificial intelligence, combinatorial optimization, databases, cryptography, networks, game theory
Scholarship: MixtAI | Honors: Highest distinction
- 2019–2022 **Double Bachelor's Degree in Biology and Computer Science**, *University of Versailles Saint-Quentin-en-Yvelines*, Versailles, France.
Computer Science: algorithms, C, Java, graph theory, databases, cryptography, networks
Biology: genetics, transcription, translation, phylogeny, sequencing, molecular biology techniques
Honors: Highest distinction
- 2019 **Scientific Baccalaureate (High School Diploma)**, *Lycée de la Haie Griselle*, Gérardmer, France.
Major in Life Sciences | Honors: Highest distinction

Academic Responsibilities

- 2025–Present **PhD Student Representative**, *DAVID Lab., University of Versailles Saint-Quentin-en-Yvelines*, Participation in general assemblies and dissemination of information to PhD Students; assistance in drafting the laboratory's new bylaws; and help in organizing a joint seminar for doctoral students from the DAVID laboratory and the CRYPTO team at the LMV .

Employment and Internships

- Apr–Sep 2024 **Master's Internship**, *Laboratoire DAVID, UVSQ*, Versailles, France.
In silico generation of organic molecular cages around target molecules. Path construction between binding motifs. Algorithms, molecular modeling, C language
Supervisors: Yann Strozecki, Sandrine Vial
- Mar–Jun 2024 **Academic Tutoring**, *Saclay AI School, UVSQ*, Versailles, France.
Tutoring undergraduate computer science students
- Jun–Jul 2023 **Research Internship**, *Laboratoire DAVID, UVSQ*, Versailles, France.
Development of construction guides for organic molecular cages. Improvement of an existing C program and implementation of the A* algorithm. Team-based project
Supervisors: Yann Strozecki, Sandrine Vial
- 2022–2024 **Student Ambassador**, *Université Paris-Saclay*.
Presentation of academic programs at orientation fairs. Interaction with prospective students and families
- 2022 **Research Engineer (Freelance Status)**, *CHCSC, UVSQ*.
Development of data analysis programs in R, Shiny applications (GitHub), Docker usage (96-hour mission)
- Jun–Aug 2022 **Research Internship**, *CHCSC, UVSQ*.
Development of a data analysis method based on double seriation
Supervisor: Guillaume Sapriel
- 2021–2022 **Research Internship**, *Atelier de Bio-Informatique, ISYEB – MNHN*, Paris, France.
Double seriation data analysis applied to genomic datasets. R language
Supervisor: Guillaume Sapriel

Research Activities

International Conferences

- 2026 Noé Demange, Yann Strozecki, and Sandrine Vial. Computational Generation of Substrate-Specific Molecular Cages. In Martin Aumüller and Irene Finocchi, editors, *24th International Symposium on Experimental Algorithms (SEA 2026)*, volume 371 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 15:1–15:21, Dagstuhl, Germany, 2026. Schloss Dagstuhl – Leibniz-Zentrum für Informatik. <https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.SEA.2026.15>.

Preprint

- 2026 Noé Demange and Yann Strozecki. Complexity of finding and enumerating interconnection trees, 2026. <https://arxiv.org/abs/2605.18125>.

Posters

- 2025 **MaDICS Symposium.**
Generation of construction guides for molecular cages
- 2025 **Université Paris-Saclay Research Days.**
Generation of construction guides for molecular cages

Talks

- 2026 **PhD Roundtable**, *Faculty of Sciences, UVSQ – Paris-Saclay University*, Versailles.
Presentation entitled "What Is a PhD Thesis?" for first-year undergraduate students in Mathematics–Physics–Chemistry, as part of the PPEI – Personal Study and Career Development Project
- 2024 **PhD Students Day**, *Laboratoire DAVID*.

Academic Service

- 2025 **Junior Program Committee Member**, *Junior Conference on Informatics: Theory and Applications (JITA)*.
Conference organization, call for papers, communication, peer reviewing

Supervision

Internships :

- Elouan Cossec (L3, 2026), Creation of an interface and quality criteria for the construction of molecular cages.
Supervisors : Noé Demange, Yann Strozecki, Sandrine Vial
- Nilsea Hannah (L3, 2026), Algorithms and implementation to guide path generation for molecular cage construction.
Supervisors : Noé Demange, Yann Strozecki, Sandrine Vial
- Lysa Percevault (L3, 2026), Algorithms and implementation to optimize path generation for molecular cage construction.
Supervisors : Noé Demange, Yann Strozecki, Sandrine Vial
- Elouan Cossec et Lysa Percevault (L2, 2025), Generation of Binding Motifs for Molecular Cage Construction.
Supervisors : Noé Demange, Yann Strozecki, Sandrine Vial

Master's Project (M1):

- Hajar Cherqi et He Yu (2026), Graph Drawing.
Supervisors : Noé Demange, Chloé Godet
- Raphael Dupuy, Quentin Soumet, Cyriac Thibaudeau et Théotime Turmel (2025), Pathfinding.
Supervisors : Noé Demange, Yann Strozecki

Teaching Experience

- 2026 **Foundations of Computer Science II – Python Programming**, *List, dictionary, GUI, file management, hash table*, UVSQ.
Undergraduate level (Year 1), 24h TD, Course coordinator: Mustapha Lebbah
- 2026 **Algorithms and Foundations for Data Science**, *Statistics, Clustering, PCA*, UVSQ.
Undergraduate level (Year 3), 39h TD, Course coordinator: Sandrine Vial
- 2025 **Models of Computation and Complexity**, *Pushdown automaton, Turing machine, other computational models, decidability, reduction, complexity class*, UVSQ.
Undergraduate level (Year 3), 27h TD, Course coordinator: Yann Strozecki
- 2024 **Data Structures and Algorithms**, *Recursion, sorting, stacks/queues, trees, algorithms, and complexity*, UVSQ.
Undergraduate level (Year 2), 36h TD, Course coordinator: Sandrine Vial

Academic Projects

- Oct 2023–Feb 2024 **Master's Project (M2)**, *AMIS Master's Program*.
Construction of cycle graphs of molecules and pairwise comparison using MCIS. Algorithms, molecular modeling, C, Python, R. Team of 4
Supervisors: Franck Quessette, Sandrine Vial
- Jan–May 2023 **Master's Project (M1)**, *AMIS Master's Program*.
Generation of construction guides for organic molecular cages. Reuse and extension of previous PhD and internship code. C language. Team of 5
Supervisor: Sandrine Vial
- Jan–May 2022 **Bachelor's Project (L3)**, *Double Degree in Biology and Computer Science*.
Implementation of the algorithm described in *A Dynamic Programming Algorithm for RNA Structure Prediction Including Pseudoknots* (Rivas & Eddy). C language. Team of 4
Supervisor: Sandrine Vial